ALEX MORGAN

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US DoD Secret Clearance (Interim TS/SCI Eligible)

PROFESSIONAL SUMMARY

Senior Systems & Network Software Engineer with 7+ years of experience designing, deploying, and hardening distributed space-ground communication interfaces, Software-Defined Networks (SDN), and mission-critical APIs. Proven background delivering high-reliability integration pipelines for NASA, DoD, and National Security systems. Specialized in low-latency LEO satellite constellation routing architectures, automated space safety protocols, and resilient test simulation environments.

TECHNICAL SKILLS & CORE COMPETENCIES

Networking & Topology: Software-Defined Networking (SDN), Network Topology, OpenFlow, BGP, LEO Inter-Satellite Links (ISL), CCSDS Protocols, Telemetry & Telecommand (TM/TC), Latency Optimization.

APIs & Systems: High-Availability Distributed Systems, RESTful APIs, gRPC, Protobuf, Kafka, Fault-Tolerant Microservices.

Languages & Cloud: Python, Go, C++, Rust, FastAPI, Docker, Kubernetes, Linux Kernel Tuning, AWS GovCloud.

Security & Simulation: Zero Trust, NIST SP 800-53, FIPS 140-3, DoD IL5/IL6, Network Emulation (Mininet, CORE), Orbital Simulation (Orekit).

PROFESSIONAL EXPERIENCE

Senior Software & Network Engineer — Orbital Dynamics & Space Systems

Jun 2022 – Present | Redmond, WA

- Architected and launched secure gRPC and RESTful interface APIs to link Low Earth Orbit (LEO) satellite constellation telemetry to US Government networks (NASA, DoD).
- Applied deep network topology insights and Software-Defined Networking (SDN) principles to coordinate dynamic multi-beam routing, maintaining 99.999% availability under variable links.
- Created mission-critical space safety services automating conjunction risk assessment and collision avoidance maneuvers using real-time orbital ephemeris data.
- Built an end-to-end simulated hardware-in-the-loop testbed to validate architecture stability, interoperability, and throughput prior to operational release.
- Communicated system health metrics, risk assessments, and escalation items across multi-disciplinary engineering and senior government customer boards.

Network Systems Engineer — Defense Network Solutions Inc.

Aug 2019 – May 2022 | Reston, VA

- Engineered robust interface gateways facilitating rapid, reliable communication between edge sensor platforms and secure customer command backbones.
- Ensured strict system adherence to US Government and DoD security, availability, upgradeability, and interoperability criteria (NIST 800-53, DISA STIGs).
- Partnered closely with internal product teams and external aerospace vendors to deliver custom software interface requirements ahead of schedule.
- Resolved high-severity incidents rapidly by identifying network bottlenecks and implementing resilient automated failover mechanisms.

Software Engineer — AeroTech Communications

Jun 2017 – Jul 2019 | Denver, CO

- Designed and maintained network diagnostic APIs and visual telemetry dashboards for ground-station communication links.
- Constructed automated Python-based test suites and packet load generators, simulating multi-node mesh topologies to assess transport layer performance.

SELECTED PROJECTS

ConstellationSim Engine: Built a containerized LEO network simulation test harness in Go/Python modeling Doppler shifts, dynamic packet routing, and ground-station handoffs.

Space Safety Gateway: Developed an open telemetry translation pipeline converting proprietary flight data into

EDUCATION & CERTIFICATIONS

B.S. in Computer Science & Network Engineering
University of Washington, Seattle, WA

Certifications
• Certified Information Systems Security Professional (CISSP)